Weak-* closability of the Fourier gradient without the approximation property

Candidate full solution to Remark 3.13 of arXiv:1903.10151

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Status. `candidate full solution (likely valid; human review requested)`.

Source problem

Arhancet and Kriegler [1] define, for a discrete group G, a real one-cocycle $b_\psi : G \to H$, and $-1 \leq q < 1$, the Fourier gradient

$$\partial_{\psi,q} : \mathcal{P}_G \subseteq \mathrm{VN}(G) \longrightarrow \Gamma_q(H) \rtimes_\alpha G, \qquad \partial_{\psi,q}(\lambda_s) = s_q(b_\psi(s)) \rtimes \lambda_s.$$

Their Proposition 3.12 (PDF pages 53–54) proves weak-* closability when G has the approximation property (AP). Remark 3.13 (PDF page 54) states: "We do not know if the assumption 'AP' in Proposition 3.12 is really necessary." The result below removes AP completely.

Transcription of the question

Is the AP assumption in Proposition 3.12 necessary for weak-* closability of $\partial_{\psi,q}$?

Source evidence

Proposition 3.12 and its weak-* closability definition (source PDF page 53).

By (2.1), we deduce (3.32). ■

For the property AP in the next proposition, we refer to the preliminary Subsection 2.4.

Proposition 3.12 *Assume $-1 \leqslant q < 1$. Let G be a discrete group with* AP. *The operator $\partial_{\psi,q} \colon \mathcal{P}_G \subseteq \mathrm{VN}(G) \to \Gamma_q(H) \rtimes_\alpha G$ is weak* closable*[26]. *We denote by $\partial_{\psi,q,\infty}$ its weak* closure.*

26. That is, if (x_n) is a sequence in $\mathcal{P}_G$ such that $x_n \to 0$ and $\partial_{\psi,q}(x_n) \to y$ for some $y \in \Gamma_q(H) \rtimes_\alpha G$ both for the weak* topology, then $y = 0$.

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End of the source proof and Remark 3.13 (source PDF page 54).

Proof : Suppose that (x_i) is a net of $\mathcal{P}_G$ which converges to 0 for the weak* topology with $x_i = \sum_{s \in G} x_{i,s} \lambda_s$ such that the net $(\partial_{\psi,q}(x_i))$ converges for the weak* topology to some y belonging to $\Gamma_q(H) \rtimes_\alpha G$. Let (M_{φ_j}) be the net of Fourier multipliers approximating the identity from Proposition 2.21. For any j, we have

$$\partial_{\psi,q}(M_{\varphi_j} x_i) = \partial_{\psi,q}\Bigg(M_{\varphi_j}\bigg(\sum_{s \in G} x_{i,s}\lambda_s\bigg)\Bigg) = \partial_{\psi,q}\bigg(\sum_{s \in \operatorname{supp}\varphi_j} x_{i,s}\lambda_s\bigg) = \sum_{s \in \operatorname{supp}\varphi_j} x_{i,s}\partial_{\psi,q}(\lambda_s)$$

$$= (\mathrm{Id} \rtimes M_{\varphi_j})\partial_{\psi,q}\bigg(\sum_{s \in G} x_{i,s}\lambda_s\bigg) = (\mathrm{Id} \rtimes M_{\varphi_j})\partial_{\psi,q}(x_i) \xrightarrow[i]{} (\mathrm{Id} \rtimes M_{\varphi_j})(y).$$

On the other hand, for all $s \in \operatorname{supp}\varphi_j$ we have $x_{i,s} \to 0$ as $i \to \infty$. Hence

$$\partial_{\psi,q}(M_{\varphi_j} x_i) = \partial_{\psi,q}\Bigg(\sum_{s \in \operatorname{supp}(\varphi_j)} \varphi_j(s) x_{i,s}\lambda_s\Bigg) \overset{(2.70)}{=} \sum_{s \in \operatorname{supp}(\varphi_j)} \varphi_j(s) x_{i,s} s_q(b_\psi(s)) \rtimes \lambda_s \xrightarrow[i]{} 0.$$

This implies by uniqueness of the limit that $(\mathrm{Id} \rtimes M_{\varphi_j})(y) = 0$ for any j. By the point 3 of Proposition 2.21, we deduce that $y = \mathrm{w}^*\text{-}\lim_j (\mathrm{Id}_{\Gamma_q(H)} \rtimes M_{\varphi_j})(y) = 0$. ■

Remark 3.13 We do not know if the assumption "AP" in Proposition 3.12 is really necessary.

The following generalizes an observation of [JMP2] (in the case $q = 1$).

Proof intuition

The gradient is diagonal in the group Fourier variable. Instead of using AP to approximate a possible weak-* limit by finite-support multipliers, extract each crossed-product Fourier coefficient directly. Coefficient extraction is normal, every coefficient of a graph limit above zero vanishes, and the orthogonal Fourier decomposition of the crossed-product L^2 space makes the coefficients separating.

Theorem 1. *Let G be any discrete group, let $b_\psi : G \to H$ and α be as in the source construction, and let $-1 \le q < 1$. Then*

$$\partial_{\psi,q} : \mathcal{P}_G \subseteq \mathrm{VN}(G) \longrightarrow \Gamma_q(H) \rtimes_\alpha G$$

is weak- closable. In particular, AP is unnecessary in Proposition 3.12 of [1].*

Proof. Put $N = \Gamma_q(H) \rtimes_\alpha G$, and let $E_\Gamma : N \to \Gamma_q(H)$ be the canonical normal faithful conditional expectation. For $s \in G$, define the Fourier coefficient map

$$C_s : N \longrightarrow \Gamma_q(H), \qquad C_s(z) = E_\Gamma\big(z(1 \rtimes \lambda_s)^*\big).$$

It is weak-* continuous because E_Γ and right multiplication by a fixed unitary are normal. On Fourier polynomials,

$$C_s\left(\sum_{t\in F} a_t \rtimes \lambda_t\right) = a_s.$$

Suppose that a net (x_i) in $\mathcal{P}_G$ satisfies

$$x_i \xrightarrow{\mathrm{w}^*} 0, \qquad \partial_{\psi,q}(x_i) \xrightarrow{\mathrm{w}^*} y \in N.$$

Write $x_i = \sum_t x_{i,t}\lambda_t$, where each sum is finite. For every fixed $s \in G$,

$$x_{i,s} = \tau_G(x_i\lambda_s^*) \longrightarrow 0,$$

since $x \mapsto \tau_G(x\lambda_s^*)$ is a normal functional on $\mathrm{VN}(G)$. The definition of the gradient gives

$$C_s\big(\partial_{\psi,q}(x_i)\big) = x_{i,s}s_q(b_\psi(s)).$$

By normality of C_s,

$$C_s(y) = \lim_i x_{i,s}s_q(b_\psi(s)) = 0 \qquad (s \in G).$$

Finally, crossed-product Fourier coefficients are unique. More explicitly, for the canonical finite trace,

$$L^2(N) = \bigoplus_{s\in G}^{\perp} L^2(\Gamma_q(H))(1 \rtimes \lambda_s),$$

and $C_s(y)$ is the s-th coordinate of y. Since the bounded element $y \in N$ lies in $L^2(N)$ and all its coordinates vanish, $y = 0$. This is weak-* closability. The argument works for nets and therefore also for the sequential formulation in footnote 26 of [1]. □

Verification and limitations

The only structural inputs are the normal canonical expectation, the explicit gradient formula, and uniqueness of Fourier coefficients. The coefficient convention has been checked directly on $a \rtimes \lambda_t$. No convergence of the full Fourier series in operator norm is assumed. The range $-1 \le q < 1$ is retained: at $q = 1$ the classical Gaussian field operator is generally unbounded, so this is not the same von-Neumann-algebra-valued map. The theorem resolves Remark 3.13 only; it does not address the later endpoint Lipschitz-algebra or optimal commutator questions.

Novelty check

A bounded search on 11 August 2026 covered the run indexes, the exact remark sentence and label, the source title and arXiv id, combinations of "weak-* closable", "Γ_q", "crossed product gradient", and "AP", citing-paper results, and the public record of the authors' 2022 Springer Lecture Notes volume. No later resolution of Remark 3.13 was found. This supports but does not establish novelty.

Human-review recommendation

Verify the convention for C_s and the standard orthogonal L^2 Fourier decomposition. Once those two points are confirmed, the proof has no further dependency.

References

[1] C. Arhancet and C. Kriegler, *Riesz transforms, Hodge–Dirac operators and functional calculus for multipliers I*, arXiv:1903.10151v9 (2020), Remark 3.13; arXiv:1903.10151.